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On the rate of growth of Bessel–Fourier coefficients for integrable functions

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ABSTRACT

It is known that the Bessel–Fourier coefficients $f_{\nu,m}$ of a function f, such that $x^{s+1/2}f(x)$ is integrable over $[0,1]$ with $0 \leqslant s \leqslant \nu + 1/2$, satisfy $j_{\nu,m}^{-s-1/2} f_{\nu,m} \to 0$. We show a partial converse, namely that for $0 \leqslant \alpha < 1/2$ and any non-negative $a_m \to 0$, there is a function f such that $x^{\alpha+1+\nu}f(x)$ is integrable and its Bessel–Fourier coefficients $f_{\nu,m}$ satisfy $j_{\nu,m}^{-\alpha} f_{\nu,m} \geqslant a_m$ and $j_{\nu,m}^{-\alpha} f_{\nu,m} \to 0$. For $\nu = 0$, we conjecture that the same should be true when $\alpha = \frac{1}{2}$, and discuss some consequences of this conjecture for the divergence Bessel–Fourier expansions of radial functions on the disc.

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1. Introduction

The Riemann–Lebesgue Lemma is a classic result in Fourier analysis, which tells us that the Fourier coefficients of an integrable function, f, decay to 0 at high frequencies, i.e. $\widehat{f}(n) \to 0$ as $|n| \to \infty$.

By increasing the regularity of f, say $f \in C^k$, we can obtain faster rates of decay of the coefficients, *viz.* $|n|^k \widehat{f}(n) \to 0$. But if we merely assume that $f \in L^1$, then the Riemann–Lebesgue Lemma is the best possible result we can expect. Indeed, we can find integrable functions whose Fourier coefficients decay as slowly as we like, as the following theorem shows—see, e.g., [2, Theorem 3.3.4].

Theorem 1.1. *Let $a_n \geqslant 0$ be a sequence such that $a_n \to 0$ as $|n| \to \infty$. Then, there is a function $f \in L^1([0,1])$ whose Fourier coefficients satisfy $\widehat{f}(n) \geqslant a_n$ for all $n \in \mathbb{Z}$.*

This may seem 'obvious': if we have the coefficients a_n, can we not just set

$$f(x) = \sum_{n=-\infty}^{\infty} a_n e^{2\pi i n x}?$$

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The answer is no, since there is no guarantee that this series will converge to an L^1 function. Recall that the *Fejér kernel* is given by

$$F_N(x) = \sum_{n=-N}^{N} \left(1 - \frac{|n|}{N+1}\right) \mathrm{e}^{2\pi i n x} \equiv \frac{1}{N+1} \frac{\sin(\pi(N+1)x)^2}{\sin(\pi x)^2}, \tag{1.1}$$

and consider the function

$$f(x) := \sum_{m=0}^{\infty} \frac{1}{2^m} F_{2^{2^{2^m}}}(x).$$

Since $F_N(x) \geqslant 0$ and $\|F_N\|_{L^1([0,1])} = 1$ for all N, this function is clearly well-defined and integrable. However, as Exercises 4.2.3 and 4.2.4 in [2] show, the partial sums of the Fourier series of f diverge (in the L^1 norm) like $N \log(N)$.

Note that the Fourier coefficients of f are the nasty-looking

$$\widehat{f}(n) = \sum_{m=\lceil \log_2 \log_2 \log_2(|n|) \rceil}^{\infty} \left(\frac{1}{2^m} - \frac{|n|}{2^{2^{2^n}} + 1}\right),$$

which nevertheless tend to 0 as $|n| \to \infty$ by the Riemann–Lebesgue Lemma.

In some situations, Fourier series may not be the ideal choice to expand our functions. For instance, when working on the disc

$$\mathbb{D} := \{(r\cos(2\pi\theta), r\sin(2\pi\theta)) : r, \theta \in [0,1]\},$$

it is convenient to expand $f\colon \mathbb{D} \to \mathbb{C}$ using eigenfunctions of the Laplacian on $\mathbb{D}$. This leads to the (2-dimensional) *Bessel–Fourier series* of f,

$$\sum_{n=-\infty}^{\infty} \sum_{m=1}^{\infty} f_{n,m} J_{|n|}(j_{|n|,m} r) \mathrm{e}^{2\pi i n\theta}, \tag{1.2}$$

where J_n are Bessel functions of the first kind (of order $n \in \mathbb{N}$), $j_{n,m} > 0$ ($m \in \mathbb{N}$) are the zeros of J_n, and $f_{n,m}$ are the *Bessel–Fourier coefficients*

$$f_{n,m} := \frac{2}{J_{|n|+1}(j_{|n|,m})^2} \int_0^1 \int_0^1 f(r,\theta) J_{|n|}(j_{|n|,m} r) \mathrm{e}^{2\pi i n\theta} r \,\mathrm{d}r \,\mathrm{d}\theta.$$

Note the $r\,\mathrm{d}r$ in the measure coming from the use of polar coordinates on $\mathbb{D}$.

If f is radial, i.e. $f(r,\theta) \equiv f(r)$ independently of θ, then the only non-zero (angular) Fourier coefficient occurs for $n = 0$. In this manner, we obtain a one-dimensional Bessel–Fourier series for f of order zero:

$$\sum_{m=1}^{\infty} f_m J_0(j_m x), \tag{1.3}$$

where we have set $j_m := j_{0,m}$ for notational convenience, and the Bessel–Fourier coefficients now take the form

$$f_m := \frac{2}{J_1(j_m)^2} \int_0^1 f(x) J_0(j_m x) x \,\mathrm{d}x.$$

In fact, Bessel functions (of the first kind) are defined for all complex indexes ν with $\operatorname{Re}\nu > -1$. For real ν, we have the following 'Riemann–Lebesgue Lemma' for the coefficients $f_{\nu,m}$ as follows [7].

Theorem 1.2. *Fix $0 \leqslant s \leqslant \nu + \frac{1}{2}$. If $x^{1/2+s} f(x) \in L^1([0,1])$ and $f_{\nu,m}$ are the Bessel–Fourier coefficients of order $\nu > -\frac{1}{2}$ for f, then*

$$\lim_{m\to\infty} \frac{f_{\nu,m} J_\nu(j_{\nu,m}x)}{j_{\nu,m}^s} = 0 \quad \textit{uniformly in} \quad x \in [0,1].$$

This can be proved by recalling that $J_\nu(x)$ behaves asymptotically like $\frac{1}{\sqrt{x}}\cos(x)$; the $\sqrt{x}$ is controlled by the condition $x^{1/2+s} f(x) \in L^1$, and one can apply the classical Riemann–Lebesgue Lemma for cosines to the function $x^s f(x)$. In fact, for $\nu \neq 0$, taking $x > 0$ in Theorem 1.2 yields the rates

$$\lim_{m\to\infty} \frac{f_{\nu,m}}{j_{\nu,m}^s} = 0 \quad (\nu \neq 0);$$

while, for $\nu = 0$, we have $J_0(0) = 1$, so taking $x = 0$ yields the stronger result

$$\sqrt{x} f \in L^1 \implies \lim_{m\to\infty} f_{0,m} = 0.$$

Of particular interest are the following special cases:

$$\sqrt{x} f(x) \in L^1 \implies \lim_{m\to\infty} \frac{f_{\nu,m}}{\sqrt{j_{\nu,m}}} = 0$$

$$x^{\nu+1} f(x) \in L^1 \implies \lim_{m\to\infty} \frac{f_{\nu,m}}{j_{\nu,m}^{\nu+1/2}} = 0.$$

In light of Theorem 1.2, we might suspect that we can find L^1 functions whose coefficients can be made to blow up at any given rate below $\sqrt{j_m}$ (whenever $\sqrt{x} f(x) \in L^1$) or even j_m (if $xf(x) \in L^1$). The following example supports our suspicions.

Example 1.3. Consider the functions $f^{(p)}(x) = x^{-p}$ for $p > 0$, so that $xf^{(p)}(x) \in L^1$ if, and only if, $p < 2$. Taking $p = 2 - \varepsilon$, the functions $f^{(p)}$ are in $L^1(x\,\mathrm{d}x)$ for all $\varepsilon > 0$. Their Bessel–Fourier coefficients can be estimated as follows (where '$\approx$' is a shorthand for 'neglecting constants'):

$$\begin{aligned} f_m^{(p)} &= \frac{2}{J_{\nu+1}(j_{\nu,m})^2} \int_0^1 x^{-2+\varepsilon} J_\nu(j_{\nu,m}x) x\,\mathrm{d}x \\ &\approx j_{\nu,m} \int_0^{j_{\nu,m}} j_{\nu,m}^{1-\varepsilon} \frac{J_\nu(y)}{y^{1-\varepsilon}} \frac{\mathrm{d}y}{j_{\nu,m}} \\ &\approx j_{\nu,m}^{1-\varepsilon} \left(\int_0^1 \frac{J_\nu(y)}{y^{1-\varepsilon}}\,\mathrm{d}y + \int_1^{j_{\nu,m}} \frac{\cos(y - \frac{\pi}{4} - \frac{\pi\nu}{2})}{y^{3/2-\varepsilon}}\,\mathrm{d}y + \int_1^{j_{\nu,m}} O(y^{-5/2+\varepsilon})\,\mathrm{d}y \right) \end{aligned}$$

The first integral is clearly finite; the third is clearly bounded uniformly in $j_{\nu,m}$. Finally, one can control the second integral by integrating by parts and using the asymptotic expansion for the zeros $j_{\nu,m} = \pi m - \pi/4 + \pi\nu/2 + O(1/m^3)$—see Lemma 2.2. Therefore, $f_{\nu,m}^{(\alpha)} = O(j_{\nu,m}^{1-\varepsilon})$.

In particular, taking $\varepsilon = \frac{1}{2}$, we have $f^{(1/2)}(x) = x^{-3/2}$, so that $xf^{(1/2)}(x) \in L^1$ but $\sqrt{x}f^{(1/2)}(x) \notin L^1$. In this case, the coefficients grow as $\sqrt{j_{\nu,m}}$. This example shows that $\sqrt{j_{\nu,m}}$ is a 'limiting rate of growth' for functions in $L^1(\sqrt{x}\,\mathrm{d}x)$.

The decay of Bessel–Fourier coefficients has been exploited recently by Saadi and Daher [4,5] to study when a function belongs to a Lipschitz class.

In this paper, we prove the following analogue of Theorem 1.1 for Bessel–Fourier series, which is a partial converse to Theorem 1.2.

Theorem 1.4. *Let $\nu > -\frac{1}{2}$, $0 \leqslant \alpha < \frac{1}{2}$ and*

$$\beta(\nu,\alpha) := \begin{cases} \alpha+\nu+1, & \text{if } \nu \geqslant 0; \\ \alpha+1, & \text{if } -\frac{1}{2} < \nu < 0 \text{ and } \alpha+\nu \geqslant 0; \\ 1-\nu, & \text{if } -\frac{1}{2} < \nu < 0 \text{ and } \alpha+\nu < 0. \end{cases}$$

Consider a sequence $a_m \geqslant 0$ with $a_m \to 0$. Then, there is a function $f \in L^1(x^{\beta(\nu,\alpha)}\,\mathrm{d}x)$, whose Bessel–Fourier coefficients, $f_{\nu,m}$, satisfy

$$0 \leqslant a_m \leqslant \frac{f_{\nu,m}}{j_{\nu,m}^{\alpha}} \to 0 \quad \text{as} \quad m \to \infty.$$

Furthermore, f can be chosen so that $f_{\nu,2m} = 0$ for all $m \in \mathbb{N}$, or so that $f_{\nu,2m+1} = 0$ for all $m \in \mathbb{N}$.

Note that, for $\nu \geqslant 0$, Theorem 1.4 does not prove that Theorem 1.2 is 'sharp'. Indeed,

- in Theorem 1.2 we need only assume that $x^{s+1/2}f(x) \in L^1$ to get a decay of $f_{\nu,m} = o(j_{\nu,m}^{s+1/2})$, for all $0 \leqslant s \leqslant \nu + \frac{1}{2}$;
- in Theorem 1.4 we construct an f with $x^{\alpha+\nu+1}f(x) \in L^1$, with a decay of $f_{\nu,m} = o(j_{\nu,m}^{\alpha})$ for $0 \leqslant \alpha < \frac{1}{2}$.

When $\nu = 0$, the ranges for s and α match, and we see that our result requires an extra $\sqrt{x}$, but obtains a stronger decay in the $f_{0,m}$, although it leaves the end-point $s = \alpha = \frac{1}{2}$ open. This 'gap' increases as $\nu \to \infty$, as the restriction $\alpha < \frac{1}{2}$ arises from need that certain series converge, which are *independent* of ν.

When $\nu = 0$, assuming that a kernel analogous to the classical Fejér kernel F_N is positive, we can show a stronger result, namely, that for any $0 \leqslant \alpha \leqslant \frac{1}{2}$, there is an f with $\sqrt{x}f(x) \in L^1$ whose coefficients decay as $f_{0,m} = o(j_{0,m}^{\alpha})$, which is in line with Theorem 1.2 for $s = 0$. This assumption is supported by numerical evidence.

This special case $\nu = 0$ is related to the Bessel–Fourier expansions of radial functions on the disc. In Section 5, we show how to construct an L^1 function on the disc whose Bessel–Fourier series diverges in norm for any possible truncation of the partial sums, assuming that Theorem 1.4 can be improved to coefficients with $a_m = o(\sqrt{j_{0,m}})$. Here, the condition $f_{2m+1} = 0$ will be crucial.

The paper is organised as follows: in Section 2 we record some necessary facts about Bessel functions and their zeros. In Section 3, we give a rigorous proof of Theorem 1.4, without assuming any additional results; Section 4 contains a proof of a technical, but crucial lemma used in the proof of Theorem 1.4. Finally, in Section 5 we formulate and prove a stronger version of Theorem 1.4 for $\nu = 0$, assuming a plausible conjecture. We conclude with a discussion of how these methods could be applied to the construction of an L^1 function on the disc whose Bessel–Fourier series diverges regardless of how one chooses to truncate the partial sums.

Notation Throughout, we will employ the following asymptotic notation:

$$\begin{aligned} f(x) \lesssim g(x) &\iff f(x) \leqslant Cg(x) \text{ for some constant } C > 0 \\ f(x) \approx g(x) &\iff f(x) \lesssim g(x) \quad \text{and} \quad g(x) \lesssim f(x) \\ f(x) = O(g(x)) &\iff |f(x)| \lesssim g(x) \\ a_m = o(b_m) &\iff \frac{a_m}{b_m} \to 0 \quad \text{as} \quad m \to \infty. \end{aligned}$$

We adopt the convention whereby $\mathbb{N}$ is the set of *positive* integers, with $\mathbb{N}_0 := \mathbb{N} \cup \{0\}$.

2. Bessel functions and their asymptotics

In order to prove Theorem 1.4, we will need the following facts about the Bessel functions J_ν.

Lemma 2.1. *Let $\nu > -\frac{1}{2}$. For any $r > 0$, we have*

$$J_\nu(r) = \sqrt{\frac{2}{\pi}} \frac{\cos(r - \pi/4 - \pi\nu/2)}{\sqrt{r}} + R_\nu(r), \tag{2.1}$$

where

$$|R_\nu(r)| \lesssim r^\nu \int_0^2 \mathrm{e}^{-rt} t^{\nu+\frac{1}{2}} \,\mathrm{d}t + r^\nu \int_2^\infty \mathrm{e}^{-rt} t^{2\nu-1} \,\mathrm{d}t. \tag{2.2}$$

Furthermore,

$$|R_\nu(r)| \lesssim r^{-3/2} \quad \textit{for} \quad r \geqslant 1. \tag{2.3}$$

These are well-known results; see, for instance, Appendix B in [2].

We will also use the following information about the zeros of J_ν, and how they interact with $J_{\nu+1}$.

Lemma 2.2. *Let $\nu > -\frac{1}{2}$ and denote by $j_{\nu,m}$ the mth (positive) zero of J_ν. Then*

$$j_{\nu,m} = \pi m - \frac{\pi}{4} + \frac{\pi\nu}{2} + \frac{1}{8\pi m - 2\pi} + O\left(\frac{1}{m^3}\right). \tag{2.4}$$

In particular,

$$j_{\nu,m} > m \quad \textit{for all} \quad m \geqslant 1. \tag{2.5}$$

Furthermore,

$$J_{\nu+1}(j_{\nu,m}) = \sqrt{\frac{2}{\pi}} \frac{(-1)^{m+1}}{\sqrt{j_{\nu,m}}} + O\left(m^{-3/2}\right). \tag{2.6}$$

Proof. Formula (2.4) is McMahon's asymptotic expansions for large zeros (see p. 505 of [6]). On the other hand, (2.5) is easily obtained from the following inequality [3, Theorem 3]

$$j_{\nu,m} \geqslant \pi m - \frac{\pi}{4} + \frac{\pi\nu}{2} \quad \text{for all} \quad m \geqslant 1.$$

The proof of (2.6) follows from (2.4) and the asymptotic expansion (2.1) for $J_{\nu+1}$, to wit

$$J_{\nu+1}(r) = \sqrt{\frac{2}{\pi}}\frac{\cos(r - 3\pi/4 - \pi\nu/2)}{\sqrt{r}} + O(r^{-3/2}) \quad \text{for all} \quad r \geqslant 1,$$

which is valid for $r = j_{\nu,m}$ by (2.5). Writing $\varepsilon_m = j_{\nu,m} - \pi m + \frac{\pi}{2}\left(\frac{1}{2} - \nu\right) = O(m^{-1})$ and expanding

$$\cos(j_{\nu,m} - 3\pi/4 - \pi\nu/2)) = \cos(\pi(m-1) + \varepsilon_m) = (-1)^{m+1} - \frac{(-1)^m}{2}\varepsilon_m^2 + O(\varepsilon_m^4),$$

we easily deduce (2.6). □

Finally, we record two useful formulas, which will be needed in some calculations. The first is

$$J_0'(x) = -J_1(x) \quad \text{for all} \quad x \in \mathbb{R}, \tag{2.7}$$

and can be found in Appendix B to [2]. The second is

$$\int_0^{j_m} J_0(y) y \, \mathrm{d}y = j_m J_1(j_m) \quad \text{for all} \quad m \in \mathbb{N}. \tag{2.8}$$

To see this, recall that J_0 satisfies Bessel's differential equation:

$$x^2 J_0''(x) + x J_0'(x) + x^2 J_0(x) = 0,$$

and that $J_0'(x) \equiv -J_1(x)$. Integrating the differential equation therefore leads us to Eq. (2.8).

3. Proof of the main theorem

In order to construct a function whose Bessel–Fourier coefficients have the desired decay properties, we will follow the argument of Theorem 3.3.4 in [2].

Recall that a sequence $(c_m)_m$ is *convex* if

$$c_m \geqslant 0 \quad \text{and} \quad c_m + c_{m+2} \geqslant 2c_{m+1} \quad \text{for all} \quad m \in \mathbb{N}.$$

The following two lemmas are proved in Chapter 3 of [2].

Lemma 3.1 *(Lemma 3.3.2 in [2]). Given a sequence $a_m \geqslant 0$ tending to zero as $m \to \infty$, there is a convex sequence $c_m \geqslant a_m$ that also tends to zero.*

Lemma 3.2 *(Lemma 3.3.3 in [2]). Given a convex decreasing sequence $c_m \to 0$ and a fixed integer $k \geqslant 0$,*

$$\sum_{l=0}^{\infty}(l+1)(c_{l+k} + c_{l+k+2} - 2c_{l+k+1}) = c_k.$$

Let us write

$$S_{\nu,m}^{\alpha}(x) = \sum_{k=1}^{m} j_{\nu,m}^{\alpha} J_\nu(j_{\nu,k}x).$$

We will need the following 'modified Fejér kernel' for the Bessel–Fourier series:

$$K^{\alpha}_{\nu,M}(x) := \frac{1}{M+1}\sum_{m=1}^{M} S^{\alpha}_{\nu,m}(x) = \sum_{m=1}^{M}\left(1-\frac{m}{M+1}\right) j^{\alpha}_{\nu,m} J_{\nu}(j_{\nu,m}x).$$

We will also set $K^{\alpha}_{\nu,0}(x) \equiv 1$ when $M = 0$.

As we saw in Eq. (1.1), the classical Fejér kernel, $F_N(x)$, for Fourier series has a nice closed form, which shows it to be non-negative for all x and N; this makes computing L^1 norms much easier.

While we expect $K^{\alpha}_{\nu,M}$ to be positive everywhere, to prove Theorem 1.4 it suffices to obtain uniform lower bounds on $x^{\beta(\nu,m)}K^{\alpha}_{\nu,M}(x)$ for $0 \leqslant \alpha < \frac{1}{2}$.

Lemma 3.3. *Let $\nu > -\frac{1}{2}$ and $0 \leqslant \alpha < \frac{1}{2}$. Then, there is a constant $C_\nu > 0$ such that*

$$x^{\beta(\nu,\alpha)}K^{\alpha}_{\nu,M}(x) \geqslant -C_\nu \quad \textit{uniformly in} \quad M \in \mathbb{N} \quad \textit{and} \quad x \in [0,1],$$

where

$$\beta(\nu,\alpha) := \begin{cases} \alpha+\nu+1, & \text{if } \nu \geqslant 0; \\ \alpha+1, & \text{if } -\frac{1}{2} < \nu < 0 \text{ and } \alpha+\nu \geqslant 0; \\ 1-\nu, & \text{if } -\frac{1}{2} < \nu < 0 \text{ and } \alpha+\nu < 0. \end{cases}$$

Remark. Note that $\beta(\nu,\alpha) \geqslant \frac{1}{2}$ in all cases, so that $x^{\beta(\nu,\alpha)-\frac{1}{2}} \in L^{\infty}([0,1])$. This will become important later, as we consider the asymptotic expansion of $J_\nu(y) = O(y^{-1/2})$ as $y \to \infty$.

We will prove Lemma 3.3 in Section 4. Here, we will re-state and prove Theorem 1.4.

Theorem 3.4 *(= Theorem 1.4). Let $\nu > -\frac{1}{2}$, $0 \leqslant \alpha < \frac{1}{2}$ and*

$$\beta(\nu,\alpha) := \begin{cases} \alpha+\nu+1, & \text{if } \nu \geqslant 0; \\ \alpha+1, & \text{if } -\frac{1}{2} < \nu < 0 \text{ and } \alpha+\nu \geqslant 0; \\ 1-\nu, & \text{if } -\frac{1}{2} < \nu < 0 \text{ and } \alpha+\nu < 0. \end{cases}$$

Consider a sequence $a_m \geqslant 0$ with $a_m \to 0$. Then, there is a function $f \in L^1(x^{\beta(\nu,\alpha)}\,\mathrm{d}x)$, whose Bessel–Fourier coefficients, $f_{\nu,m}$, satisfy

$$0 \leqslant a_m \leqslant \frac{f_{\nu,m}}{j^{\alpha}_{\nu,m}} \to 0 \quad \textit{as} \quad m \to \infty.$$

Furthermore, f can be chosen so that $f_{\nu,2m} = 0$ for all $m \in \mathbb{N}$, or so that $f_{\nu,2m+1} = 0$ for all $m \in \mathbb{N}$.

Proof. By Lemma 3.1 there is a decreasing convex sequence $c_m \geqslant a_m$ such that $c_m \to 0$. (Note that we may need to take $c_0 = 2c_1$, say, in order to have a decreasing, convex sequence defined for $m \geqslant 0$.) We use $(c_m)_{m\geqslant 0}$ to construct the function

$$f(x) := \sum_{l=1}^{\infty}(l+1)(c_l + c_{l+2} - 2c_{l+1})K^{\alpha}_{\nu,l}(x). \tag{3.1}$$

Let $\beta = \beta(\nu,\alpha)$. We claim that $x^{\beta}f(x) \in L^1$.

First, by Lemma 3.3, $x^\beta K^\alpha_{\nu,M}(x) + C \geqslant 0$, so we have

$$\left\|x^\beta K^\alpha_{\nu,M}\right\|_{L^1([0,1])} \leqslant 2C + \int_0^1 K^\alpha_{\nu,M}(x)x^\beta \,\mathrm{d}x.$$

Writing $K^\alpha_{\nu,M}$ out as a sum, and changing variables to $y = j_{\nu,m}x$ in the integral, we have

$$\int_0^1 K^\alpha_{\nu,M}(x)x^\beta \,\mathrm{d}x = \sum_{m=1}^{M}\left(1 - \frac{m}{M+1}\right)\frac{1}{j_{\nu,m}^{\beta-\alpha+1}}\int_0^{j_{\nu,m}} J_\nu(y)y^\beta \,\mathrm{d}y. \tag{3.2}$$

We now split the integral into the ranges $0 \leqslant y \leqslant 1$ and $1 \leqslant y \leqslant j_{\nu,m}$, and use the asymptotics from Lemma 2.1:

$$\int_0^{j_{\nu,m}} J_\nu(y)y^\beta \,\mathrm{d}y = \int_0^1 J_\nu(y)y^\beta \,\mathrm{d}y \tag{3.3}$$

$$+ \int_1^{j_{\nu,m}} \sqrt{\frac{2}{\pi}}\cos(y - \pi/4 - \pi\nu/2)y^{\beta-1/2} \,\mathrm{d}y \tag{3.4}$$

$$+ \int_1^{j_{\nu,m}} O\left(y^{\beta-3/2}\right) \,\mathrm{d}y. \tag{3.5}$$

Now, for (3.3), we immediately have

$$\frac{1}{j_{\nu,m}^{\beta-\alpha+1}}\int_0^1 J_\nu(y)y^\beta \,\mathrm{d}y \lesssim \frac{1}{j_{\nu,m}^{\beta-\alpha+1}},$$

and for (3.5),

$$\frac{1}{j_{\nu,m}^{\beta-\alpha+1}}\int_1^{j_{\nu,m}} O\left(y^{\beta-3/2}\right) \,\mathrm{d}y \lesssim \frac{1}{j_{\nu,m}^{3/2-\alpha}}.$$

To get suitable decay in (3.4), we have to integrate by parts. Neglecting nuisance factors of $\frac{2}{\pi}$, we have

$$\int_1^{j_{\nu,m}} \cos\left(y - \frac{\pi}{4} - \frac{\pi\nu}{2}\right)y^{\beta-1/2} \,\mathrm{d}y = \left[\sin(y - \pi/4 - \pi\nu/2)y^{\beta-1/2}\right]_1^{j_{\nu,m}}$$

$$- \int_1^{j_{\nu,m}} \sin\left(y - \frac{\pi}{4} - \frac{\pi\nu}{2}\right)y^{\beta-3/2} \,\mathrm{d}y.$$

The second integral is trivial to bound, while from the sine term and Eq. (2.4) we obtain

$$\frac{j_{\nu,m}^{\beta-1/2}}{j_{\nu,m}^{\beta-\alpha+1}}\sin\left(j_{\nu,m} - \frac{\pi}{4} - \frac{\pi\nu}{2}\right) = \frac{(-1)^{m+1}}{j_{\nu,m}^{3/2-\alpha}} + O\left(\frac{1}{m^2}\right).$$

Since $\alpha < 1/2 \leqslant \beta$, the terms from Eqs. (3.3) to (3.5) are summable and, therefore, (3.2) converges. In other words, $\left\|x^\beta K^\alpha_{\nu,M}\right\|_{L^1([0,1])} = O(1)$ uniformly in M, and so, from Eq. (3.1) and Lemma 3.2, we see that

$$\left\|x^\beta f\right\|_{L^1([0,1])} \lesssim \sum_{l=0}^{\infty}(l+1)(c_l + c_{l+2} - 2c_{l+1}) = c_0 < \infty,$$

completing the proof of our claim.

It is now straightforward to show that

$$f_{\nu,m} = j^\alpha_{\nu,m} c_m \geqslant a_m \quad \text{and so} \quad \frac{f_{\nu,m}}{j^\alpha_{\nu,m}} = c_m \to 0.$$

A slight modification of that argument shows that we can find such an f with the additional property that $f_{2m+1} = 0$ for all $m \in \mathbb{N}$. It suffices to replace the kernel $K^\alpha_{\nu,M}$ with the 'even kernel'

$$K^{\alpha,e}_{\nu,M}(x) := \sum_{m=1}^{M}\left(1 - \frac{m}{M+1}\right) j^\alpha_{\nu,m} J_\nu(j_{\nu,2m}x).$$

As can be seen from the proof of Lemma 3.3 below, we still have that $x^\beta K^{\alpha,e}_{\nu,M}(x) \geqslant -C$ uniformly in $M \in \mathbb{N}$ and $x \in [0,1]$. We construct f as in Eq. (3.1) and show that it satisfies $x^\beta f(x) \in L^1$, just as before. Then, the same computation of the f_m shows that

$$f_{\nu,2m} = j^\alpha_{\nu,m} c_m \quad \text{and} \quad f_{\nu,2m+1} = 0 \quad \text{for all} \quad m \in \mathbb{N}, \tag{3.6}$$

as required. □

4. Bounds on $x^{\beta(\nu,\alpha)} K^\alpha_{\nu,M}(x)$

In this section, we prove Lemma 3.3. First, we will need another technical lemma.

Lemma 4.1. *For $0 < \alpha < \frac{1}{2}$ and $u > 0$,*

$$\max_{m>0}\left\{m^\alpha \mathrm{e}^{-mu}\right\} = \left(\frac{\alpha}{\mathrm{e}u}\right)^\alpha \lesssim u^{-\alpha}.$$

On to the proof of Lemma 3.3. To make things simpler, we will assume throughout that $0 < \alpha < \frac{1}{2}$; the necessary modifications for $\alpha = 0$ are straightforward. Once again, we let $\beta = \beta(\nu,\alpha)$, for convenience.

To begin, note that it is sufficient to prove that $x^\beta S^\alpha_{\nu,M}(x) \geqslant -C$ uniformly in x and M. Using the asymptotics from (2.1), we have that

$$\begin{aligned} x^\beta S^\alpha_{\nu,M}(x) &= x^\beta \sum_{m=1}^{M} j^\alpha_{\nu,m} J_\nu(j_{\nu,m}x) \\ &= \sqrt{\frac{2}{\pi}} x^{\beta-1/2} \sum_{m=1}^{M} \frac{\cos(j_{\nu,m}x - \pi/4 - \pi\nu/4)}{j^{1/2-\alpha}_{\nu,m}} \\ &\quad + \sum_{m=1}^{M} x^\beta j^\alpha_{\nu,m} R_\nu(j_{\nu,m}x) \\ &\equiv \sqrt{\frac{2}{\pi}} x^{\beta-1/2} \widetilde{S}^\alpha_{\nu,M}(x) + R^\alpha_{\nu,M}(x), \end{aligned} \tag{4.1}$$

where we have defined

$$\widetilde{S}^{\alpha}_{\nu,M} := \sum_{m=1}^{M} \frac{\cos(j_{\nu,m}x - \pi/4 - \pi\nu/2)}{j_{\nu,m}^{1/2-\alpha}} \quad \text{and} \quad R^{\alpha}_{\nu,M}(x) := \sum_{m=1}^{M} x^{\beta} j^{\alpha}_{\nu,m} R_{\nu}(j_{\nu,m}x).$$

It suffices to obtain lower bounds on $x^{\beta-1/2}\widetilde{S}^{\alpha}_{\nu,M}(x)$ and upper bounds on $\left|R^{\alpha}_{\nu,M}(x)\right|$ that are uniform in $x \in [0,1]$ and $M \in \mathbb{N}$.

Lower bounds on $\widetilde{S}^{\alpha}_{\nu,M}$ Since $x^{\beta-1/2} \geqslant 0$ and $\beta \geqslant \frac{1}{2}$, it suffices to obtain lower bounds on the sums $\widetilde{S}^{\alpha}_{\nu,M}$. This can be accomplished using an argument from [8], relating the sum to an integral. Special attention needs to be paid to the errors that accrue at the overlaps of successive intervals $[j_{\nu,m} - \frac{\pi}{2}, j_{\nu,m} + \frac{\pi}{2}]$ $(m \in \mathbb{N})$. Details for $\nu = 0$ can be found in [1], and the necessary modifications for other $\nu > -\frac{1}{2}$ are essentially notational.

Bounds on $R^{\alpha}_{\nu,M}$ We assume that $\nu \geqslant 0$, so $\beta(\nu,\alpha) = \nu + \alpha + 1$. The remaining cases, where $-\frac{1}{2} < \nu < 0$ and either $\nu + \alpha \geqslant 0$ or $\nu + \alpha < 0$, can be treated in a similar manner.

Recall that $R^{\alpha}_{\nu,M}(x) := \sum_{m=1}^{M} x^{\alpha+\nu+1} j^{\alpha}_{\nu,m} R_{\nu}(j_{\nu,m}x)$. By (2.5), we have $\mathrm{e}^{-j_{\nu,m}xt} \leqslant \mathrm{e}^{-mxt}$ for all $x \in [0,1]$ and $t \in (0,\infty)$. On the other hand, $j_{\nu,m} \lesssim m$ for all $m \geqslant 1$. Taking $u = xt/2$ in Lemma 4.1, we have

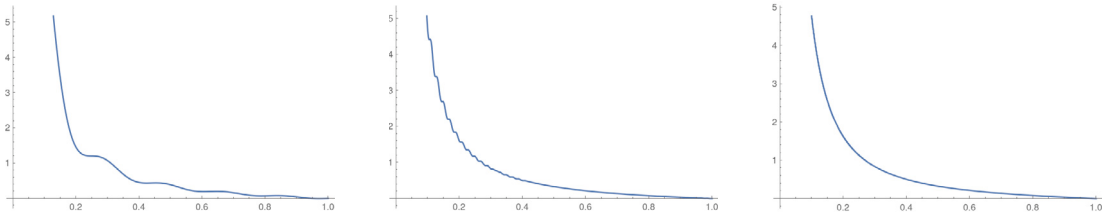
$$j^{\alpha+\nu}_{\nu,m}\mathrm{e}^{-mxt} \lesssim m^{\alpha+\nu}\mathrm{e}^{-mxt} = \frac{m^{\alpha+\nu}}{\mathrm{e}^{\frac{mxt}{2}}}\mathrm{e}^{-\frac{mxt}{2}} \lesssim \frac{\mathrm{e}^{-\frac{mxt}{2}}}{(xt)^{\alpha+\nu}}.$$

Hence, from (2.2) we obtain the following bounds for all $x > 0$:

$$\begin{aligned}
\left|R^{\alpha}_{\nu,M}(x)\right| &\leqslant x^{\alpha+2\nu+1} \sum_{m=1}^{\infty} \left[\int_0^2 m^{\alpha+\nu}\mathrm{e}^{-mxt} t^{\nu+\frac{1}{2}}\,\mathrm{d}t + x\int_2^{\infty} m^{\alpha+\nu}\mathrm{e}^{-mxt}t^{2\nu-1}\,\mathrm{d}t \right] \\
&\lesssim x^{\nu+1}\int_0^2 \sum_{m=1}^{\infty} \mathrm{e}^{-\frac{mxt}{2}} t^{\frac{1}{2}-\alpha}\,\mathrm{d}t + x\int_2^{\infty}\sum_{m=1}^{\infty}\mathrm{e}^{-\frac{mxt}{2}}t^{\nu-1-\alpha}\,\mathrm{d}t \\
&= x^{\nu+1}\int_0^2 \frac{1}{\mathrm{e}^{xt/2}-1} t^{\frac{1}{2}-\alpha}\,\mathrm{d}t + x\int_2^{\infty}\frac{1}{\mathrm{e}^{xt/2}-1}t^{\nu-1-\alpha}\,\mathrm{d}t \\
&\lesssim \int_0^2 \frac{x^{\nu+1}t^{\frac{1}{2}-\alpha}}{xt}\,\mathrm{d}t + \int_2^{\infty}\frac{xt^{\nu-1-\alpha}}{xt}\,\mathrm{d}t \\
&\lesssim x^{\nu}\int_0^2 \frac{1}{t^{\alpha+1/2}}\,\mathrm{d}t + \int_2^{\infty}\frac{1}{t^{\alpha-\nu+2}}\,\mathrm{d}t \leqslant C/2,
\end{aligned}$$

uniformly in $x > 0$ and $M \in \mathbb{N}$, since $\alpha < \frac{1}{2}$ and $\nu \geqslant 0$. Hence, we can take $x \to 0^+$ and obtain uniform bounds on $\left|R^{\alpha}_{\nu,M}(x)\right|$ for $x \in [0,1]$ and $M \in \mathbb{N}$. In conjunction with the bounds for $S^{\alpha}_{\nu,M}$, we deduce the result from (4.1).

Remark. The reader should have no trouble adapting the argument to the 'even kernel' $K^{\alpha,e}_{\nu,M}$ introduced in the proof of Theorem 1.4. Further details can be found in [1].



(a) Plot of $K^{1/2}_{0,10}(x)$. (b) Plot of $K^{1/2}_{0,100}(x)$. (c) Plot of $K^{1/2}_{0,1000}(x)$.

Fig. 1. Plots of $K^{1/2}_M(x)$ for $0 \leqslant x \leqslant 1$ with $M = 10$, $M = 100$ and $M = 1000$, respectively. (Generated with Mathematica.)

5. The case $\nu = 0$ and divergent Bessel–Fourier expansions

Numerical simulations suggest that $K^{\alpha}_{0,M}(x) \geqslant 0$ uniformly in $x \in [0,1]$ and $M \in \mathbb{N}$, for the range $0 \leqslant \alpha \leqslant \frac{1}{2}$—see Fig. 1.

We therefore make the following conjecture.

Conjecture 1. *For all $0 \leqslant \alpha \leqslant \frac{1}{2}$, $M \in \mathbb{N}$ and $x \in [0,1]$, we have $K^{\alpha}_{\nu,M}(x) \geqslant 0$.*

Assuming this conjecture, we can prove the following strengthening of Theorem 1.4.

Theorem 5.1. *Assume that Conjecture 1 is true. Let $a_m \geqslant 0$ be a sequence such that $a_m \to 0$.*

(i) If $0 \leqslant \alpha < \frac{1}{2}$, then there is a function f such that $\sqrt{x}f(x) \in L^1$ and whose Bessel–Fourier coefficients, $f_{0,m}$, satisfy

$$0 \leqslant a_m \leqslant \frac{f_{0,m}}{j^{\alpha}_{0,m}} \to 0 \quad \text{as} \quad m \to \infty.$$

Furthermore, f can be chosen so that $f_{0,2m} = 0$ for all $m \in \mathbb{N}$, or so that $f_{0,2m+1} = 0$ for all $m \in \mathbb{N}$.

(ii) If $\alpha = \frac{1}{2}$, then there is a function f such that $xf(x) \in L^1$ and whose Bessel–Fourier coefficients, $f_{0,m}$, satisfy

$$0 \leqslant a_m \leqslant \frac{f_{0,m}}{\sqrt{j_{0,m}}} \to 0 \quad \text{as} \quad m \to \infty.$$

Proof. Let $\alpha < \frac{1}{2}$ and construct f as before, using (3.1). We now claim (assuming Conjecture 1) that $\sqrt{x}f(x) \in L^1$.

By assumption (i.e. Conjecture 1), $K^{\alpha}_{0,M}(x) \geqslant 0$, so

$$\left\| \sqrt{x} K^{\alpha}_{0,M} \right\|_{L^1([0,1])} = \int_0^1 K^{\alpha}_{0,M}(x) \sqrt{x} \, \mathrm{d}x.$$

Writing $K^{\alpha}_{0,M}$ out as a sum and changing variables in the integral again, we have

$$\begin{aligned} \int_0^1 K^{\alpha}_{0,M}(x) \sqrt{x} \, \mathrm{d}x &= \sum_{m=1}^{M} \left(1 - \frac{m}{M+1}\right) j^{\alpha}_{0,m} \int_0^1 J_0(j_{0,m}x) \sqrt{x} \, \mathrm{d}x \\ &= \sum_{m=1}^{M} \left(1 - \frac{m}{M+1}\right) \frac{1}{j^{3/2-\alpha}_{0,m}} \int_0^{j_{0,m}} J_0(y) \sqrt{y} \, \mathrm{d}y. \end{aligned} \tag{5.1}$$

We now split the range of integration into two pieces. When $0 \leqslant y \leqslant 1$, we have $J_0(y)\sqrt{y} = O(1)$. When $1 \leqslant y \leqslant j_{0,m}$, we can use the asymptotics for J_0 in Lemma 2.1. Hence,

$$\frac{1}{j_{0,m}^{3/2-\alpha}} \int_0^{j_{0,m}} J_0(y)\sqrt{y}\,\mathrm{d}y = \frac{1}{j_{0,m}^{3/2-\alpha}} \int_0^{1} J_0(y)\sqrt{y}\,\mathrm{d}y \tag{5.2}$$

$$+ \frac{1}{j_{0,m}^{3/2-\alpha}} \int_1^{j_{0,m}} \sqrt{\frac{2}{\pi}} \cos\left(y - \frac{\pi}{4}\right) \mathrm{d}y \tag{5.3}$$

$$+ \frac{1}{j_{0,m}^{3/2-\alpha}} \int_1^{j_{0,m}} O\left(\frac{1}{y}\right) \mathrm{d}y. \tag{5.4}$$

Now, using the asymptotics for $j_{0,m}$ in Lemma 2.2, we have

$$(5.2) \lesssim \frac{1}{m^{3/2-\alpha}}, \quad (5.3) \lesssim \frac{1}{m^{3/2-\alpha}} \quad \text{and} \quad (5.4) \lesssim \frac{\log m}{m^{3/2-\alpha}} \lesssim \frac{1}{m^{3/2-\alpha/2}},$$

since $m^{-\alpha/2}\log m \to 0$. For $\alpha < \frac{1}{2}$, all of the above terms are summable. Combining and summing these terms in (5.1), we deduce that the norms $\left\|\sqrt{x}K^{\alpha}_{0,M}\right\|_{L^1([0,1])}$ are bounded uniformly in M. Therefore, from the definition of f in (3.1) and Lemma 3.2 we see that

$$\left\|\sqrt{x}f\right\|_{L^1([0,1])} \lesssim \sum_{l=0}^{\infty}(l+1)(c_l + c_{l+2} - 2c_{l+1}) = c_0 < \infty,$$

completing the proof of our claim.

The computation of the f_m, including the modifications to ensure that $f_{2m+1} = 0$ for all $m \in \mathbb{N}$, are identical to those in the proof of Theorem 1.4. This completes the proof of case (i).

When $\alpha = \frac{1}{2}$, we again define f using (3.1), but now integrate with respect to $x\,\mathrm{d}x$:

$$\begin{aligned}
\int_0^1 K^{1/2}_{0,M}(x)x\,\mathrm{d}x &= \sum_{m=1}^{M}\left(1 - \frac{m}{M+1}\right)\frac{\sqrt{j_{0,m}}}{j_{0,m}^2}\int_0^{j_{0,m}} J_0(y)y\,\mathrm{d}y \\
&= \sum_{m=1}^{M}\left(1 - \frac{m}{M+1}\right)\frac{\sqrt{j_{0,m}}}{j_{0,m}^2} j_{0,m}J_1(j_{0,m}) \\
&= \sum_{m=1}^{M}\left(1 - \frac{m}{M+1}\right)\frac{1}{\sqrt{j_{0,m}}}J_1(j_{0,m}) \\
&= \sqrt{\frac{2}{\pi}}\sum_{m=1}^{M}\left(1 - \frac{m}{M+1}\right)\left[\frac{(-1)^{m+1}}{j_{0,m}} + O\left(\frac{1}{m^2}\right)\right].
\end{aligned}$$

The last equation follows from (2.6), and the second sum converges as they are the Cesàro means of a convergent series. □

Remarks. We observe the following.

(i) For any $\nu > -\frac{1}{2}$, assuming that $K^{\alpha}_{\nu,M}(x) \geqslant 0$, then the proof of (i) of the theorem goes through unchanged.

(ii) Note how the oscillations coming from $\frac{J_1(j_{0,m})}{\sqrt{j_{0,m}}}$ are essential to the convergence of (3.1), and this prevents us from discarding the odd (or even) coefficients as we manage to do in case (i) of the theorem.

In light of Theorem 5.1, Conjecture 1 implies the following corollary.

Corollary 5.2. *Given any non-negative sequence $a_m \to 0$, there is an $L^1(x\,\mathrm{d}x)$ function f whose Bessel–Fourier coefficients of order 0, $f_{0,m}$, satisfy*

$$0 \leqslant a_m \leqslant \frac{f_{0,m}}{\sqrt{j_{0,m}}} \to 0 \quad \textit{as} \quad m \to \infty.$$

We conclude this note with a possible application of these results, assuming Conjecture 1, to the divergence of certain Bessel–Fourier series on the disc. To do so, we need to make an additional assumption.

Conjecture 2. *The function in Corollary 5.2 can be chosen such that*

$$f_{0,2m} = 0 \quad \textit{and} \quad f_{0,2m+1} \geqslant \sqrt{j_{0,m}} a_m \quad \textit{for all} \quad m \in \mathbb{N},$$

or with

$$f_{0,2m} \geqslant \sqrt{j_{0,2m}} a_m \quad \textit{and} \quad f_{0,2m+1} = 0 \quad \textit{for all} \quad m \in \mathbb{N}.$$

This conjecture, while a natural generalisation of Theorem 1.4 to $\alpha = \frac{1}{2}$, is less certain. Indeed, when $\alpha < \frac{1}{2}$, we were able to 'eliminate' the even (or odd) coefficients because we had enough powers of $\frac{1}{j_{0,m}}$ to ensure that the kernel $K^{\alpha}_{0,M}$ was absolutely summable in m as $M \to \infty$. But when $\alpha = \frac{1}{2}$, we have to invoke oscillations in order to retain convergence, and so both even and odd coefficients must make an appearance. A search for an alternative proof that $K^{\alpha}_{0,M}(x) \geqslant 0$ might yield a way around this 'limitation'.

Assuming both Corollary 5.2 and Conjecture 2, we can prove the following proposition.

Proposition 5.3. *Assume that Corollary 5.2 and Conjecture 2 are both true. Then, there is a function $f \in L^1(\mathbb{D})$ such that the partial sums*

$$S_{M,N}(r,\theta) = \sum_{n=-N}^{N} \sum_{m=1}^{M} c_{n,m}(f) J_{|n|}(j_{|n|,m} r) \mathrm{e}^{2\pi \mathrm{i} n\theta}$$

diverge in $L^1(r\,\mathrm{d}r\,\mathrm{d}\theta)$ for any choice of $M = M_k, N = N_k \to \infty$ as $k \to \infty$.

Proof. If the conjectures are true, then by taking $a_m = \frac{1}{\log m}$, we can find a function $g \in L^1([0,1], x\,\mathrm{d}x)$ whose 1-dimensional Bessel–Fourier coefficients satisfy

$$g_{2m} = 0 \quad \text{and} \quad g_{2m+1} \geqslant \frac{\sqrt{j_{0,m}}}{\log(m)}.$$

Set $f(r,\theta) := g(r)$, so that the 2-dimensional Bessel–Fourier series of f is given by

$$S_{M,N} f(r,\theta) = \sum_{m=1}^{M} g_m J_0(j_{0,m} r),$$

and so

$$\|S_{M,N}f\|_{L^1(\mathbb{D})} \geqslant \left| \int_0^1 S_{M,N}f(r,\theta) r \,\mathrm{d}r \right| = \left| \sum_{m=1}^{M} g_m \int_0^1 J_0(j_{0,m}r) r \,\mathrm{d}r \right|.$$

Using Eq. (2.8) we have

$$\int_0^1 S_{M,N}f(r,\theta) r \,\mathrm{d}r = \sum_{m=1}^{M} \frac{g_m}{j_{0,m}^2} \int_0^{j_{0,m}} J_0(y) y \,\mathrm{d}y = \sum_{m=1}^{M} \frac{g_m}{j_{0,m}} J_1(j_{0,m}).$$

Now, since $J_1 = -J_0'$, it is not hard to see that the signs of $J_1(j_{0,m})$ alternate, with $J_1(j_{0,2m}) < 0$ (i.e., J_0 is decreasing at $j_{0,2m}$) and $J_1(j_{0,2m+1}) > 0$ (i.e., J_0 is increasing at $j_{0,2m+1}$). Since $g_{2m} = 0$ for all m, the partial sums are non-negative and therefore, using

$$\frac{|J_1(j_{0,m})|}{\sqrt{j_{0,m}}} \gtrsim \frac{1}{j_{0,m}},$$

which follows from (2.6), we obtain

$$\|S_{N,M}f\|_{L^1(\mathbb{D})} = \sum_{m=1}^{M} \frac{g_{2m+1}}{\sqrt{j_{0,2m+1}}} \frac{|J_1(j_{0,2m+1})|}{\sqrt{j_{0,2m+1}}} \gtrsim \sum_{m=1}^{M} \frac{1}{m \log(2m+1)} \to \infty,$$

as required. □

Declaration of competing interest

The author declares no conflicts of interest.

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